

Visual

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1|      10|      20|      30|      40|      50|      60|      70|      80|      90|     100|     110|
1  R_TRIG_13 (CLK := IN_Diffuse_Sensor_16);
2  R_TRIG_14 (CLK := IN_Diffuse_Sensor_17);
3  R_TRIG_15 (CLK := IN_Diffuse_Sensor_6);
4  R_TRIG_16 (CLK := IN_Diffuse_Sensor_11);
5  R_TRIG_21 (CLK := IN_Light_Array_Emitter_1_Value > 193);
6  R_TRIG_22 (CLK := IN_Light_Array_Emitter_2_Value > 193);
7  R_TRIG_19 (CLK := IN_Diffuse_Sensor_7);
8
9  IF R_TRIG_13.Q THEN
10 Contador_azules := Contador_azules + 1;
11 OUT_Digital_Display_1 := Contador_azules;
12 END_IF;
13
14 IF R_TRIG_14.Q THEN
15 Contador_verdes := Contador_verdes + 1;
16 OUT_Digital_Display_2 := Contador_verdes;
17 END_IF;
18
19 IF R_TRIG_15.Q THEN
20 Contador_PF := Contador_PF + 1;
21 END_IF;
22
23 IF R_TRIG_16.Q THEN
24 Contador_gris := Contador_gris + 1;
25 OUT_Digital_Display_3 := Contador_gris;
26 END_IF;
27
28 IF R_TRIG_21.Q THEN
29 Contador_F_G := Contador_F_G + 1;
30 END_IF;
31
32 IF R_TRIG_22.Q THEN
33 Contador_NF_G := Contador_NF_G + 1;
34 END_IF;
35
36 IF R_TRIG_19.Q THEN
37 Contador_P_nF := Contador_P_nF + 1;
38 END_IF;
39
40
41 SC_Selector_0 := IN_Selector_1_State_0 ;
42 SC_Selector_1 := IN_Selector_1_State_1 ;
43 SC_IN_Emergency_Stop_1 := IN_Emergency_Stop_1;
44 SC_IN_Reset_Button_1 := IN_Reset_Button_1;
45 SC_IN_Start_Button_1 := IN_Start_Button_1;
46 SC_IN_Stop_Button_1 := IN_Stop_Button_1;
47
48 SC_OUT_BeltConveyor_Inclined_1 := OUT_BeltConveyor_Inclined_1;
49 SC_OUT_BeltConveyor_Inclined_2 := OUT_BeltConveyor_Inclined_2;
50 SC_OUT_BeltConveyor_Inclined_3 := OUT_BeltConveyor_Inclined_3;
51 SC_OUT_BeltConveyor_Inclined_4 := OUT_BeltConveyor_Inclined_4;
52
53 SC_OUT_BeltConveyor_4_5 := OUT_BeltConveyor_4_5;
54 SC_OUT_BeltConveyor_4_6 := OUT_BeltConveyor_4_6;
55
56 SC_OUT_Two_Axis_Pick_Place_1_X := OUT_Two_Axis_Pick_Place_1_X;
57 SC_OUT_Two_Axis_Pick_Place_1_Z := OUT_Two_Axis_Pick_Place_1_Z;
58 SC_OUT_Two_Axis_Pick_Place_2_X := OUT_Two_Axis_Pick_Place_2_X;
59 SC_OUT_Two_Axis_Pick_Place_2_Z := OUT_Two_Axis_Pick_Place_2_Z;
60
61 SC_OUT_Two_Axis_PP_1_Grab := OUT_Two_Axis_Pick_Place_1_Grab;
62 SC_OUT_Two_Axis_PP_2_Grab := OUT_Two_Axis_Pick_Place_2_Grab;
63
64 SC_OUT_Emitter4_Emit := OUT_Emitter4_Emit;
65 SC_OUT_Emitter5_Emit := OUT_Emitter5_Emit;
66
67 SC_OUT_Roller_Conveyor_V:=OUT_Roller_Conveyor_V;
68 SC_OUT_Roller_Conveyor_A:=OUT_Roller_Conveyor_A;
69 SC_OUT_Roller_Conveyor_F:=OUT_Roller_Conveyor_F;
70 SC_OUT_Roller_Conveyor_NF:=OUT_Roller_Conveyor_NF;
71
72 (* ===== *)
73 (* VARIABLES SCADA ADICIONALES - MANDO / GEMMA / ESTADO GENERAL *)
74 (* ===== *)
75
76 SC_Estado_GEMMA := Estado_GEMMA;
77 SC_Mensaje_Sistema := Mensaje_Sistema;
78
79 SC_Marcha_General := Marcha_General;
80 SC_Permiso_Proceso := Permiso_Proceso;
81 SC_Seguridad_OK := Seguridad_OK;
82 SC_Modo_AUTO := Modo_AUTO;
83 SC_Stop_Pulsado := Stop_Pulsado;
84 SC_Reset_General := Reset_General;
85 SC_Reset_Activo := OUT_FACTORY_IO_Reset;
86
87 SC_Cmd_Start := Cmd_Start;
88 SC_Cmd_Stop := Cmd_Stop;
89 SC_Cmd_Reset := Cmd_Reset;
90
91 SC_Estado_Parada_Inicial := Estado_GEMMA = 0;
92 SC_Estado_Produccion := Estado_GEMMA = 10;
93 SC_Estado_Parada_Normal := Estado_GEMMA = 20;
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1|      10|      20|      30|      40|      50|      60|      70|      80|      90|     100|     110|
94 SC_Estado_Emergencia := Estado_GEMMA = 30;
95 SC_Estado_Reset := Estado_GEMMA = 40;
96
97
98 (* ===== *)
99 (* FACTORY I/O Y CINTA PRINCIPAL *)
100 (* ===== *)
101
102 SC_FACTORY_IO_Running := IN_FACTORY_IO_Running;
103 SC_FACTORY_IO_Paused := IN_FACTORY_IO_Paused;
104 SC_FACTORY_IO_Reset := IN_FACTORY_IO_Reset;
105
106 SC_OUT_FACTORY_IO_Run := OUT_FACTORY_IO_Run;
107 SC_OUT_FACTORY_IO_Pause := OUT_FACTORY_IO_Pause;
108 SC_OUT_FACTORY_IO_Reset := OUT_FACTORY_IO_Reset;
109
110 SC_OUT_Emitter1_Emit := OUT_Emitter1_Emit;
111 SC_OUT_BeltConveyor_6_13 := OUT_BeltConveyor_6_13;
112
113
114 (* ===== *)
115 (* ALARMAS *)
116 (* ===== *)
117
118 SC_Alarma_Emergencia := NOT Seguridad_OK;
119 SC_Alarma_Sistema_Detenido := NOT Marcha_General;
120 SC_Alarma_Sin_Permiso := NOT Permiso_Proceso;
121 SC_Alarma_Parada_Normal := Estado_GEMMA = 20;
122 SC_Alarma_Reset := Estado_GEMMA = 40;
123 SC_Alarma_FIFO_Lleno := SlotCount >= 10;
124
125 SC_Alarma_Rack_Azul_Lleno := Position >= 54;
126 SC_Alarma_Rack_Verde_Lleno := Position_V >= 54;
127 SC_Alarma_Rack_F_Lleno := Position_F >= 54;
128 SC_Alarma_Rack_NF_Lleno := Position_NF >= 54;
129
130 SC_Alarma_Activa :=
131     SC_Alarma_Emergencia OR
132     SC_Alarma_Sin_Permiso OR
133     SC_Alarma_FIFO_Lleno OR
134     SC_Alarma_Rack_Azul_Lleno OR
135     SC_Alarma_Rack_Verde_Lleno OR
136     SC_Alarma_Rack_F_Lleno OR
137     SC_Alarma_Rack_NF_Lleno;
138
139
140 (* ===== *)
141 (* IDENTIFICACIÓN Y CLASIFICACIÓN *)
142 (* ===== *)
143
144 SC_IN_Sensor_Presencia_Entrada := IN_Diffuse_Sensor_1;
145 SC_IN_Sensor_Clasificacion := IN_Diffuse_Sensor_2;
146 SC_IN_Sensor_RFID := IN_Diffuse_Sensor_5;
147
148 SC_Camara_Deteccion :=
149     IN_Vision_Sensor_1_Bool_0 OR
150     IN_Vision_Sensor_1_Bool_1 OR
151     IN_Vision_Sensor_1_Bool_2 OR
152     IN_Vision_Sensor_1_Bool_3;
153
154 SC_RFID_Activo := RFID_READER_1_Executecommand OR RFID_READER_2_Executecommand;
155 SC_RFID_Read_Value := RFID_Read_Value;
156 SC_Valor_Capturado := Valor_Capturado;
157
158 SC_SlotCount := SlotCount;
159 SC_SlotRead := SlotRead;
160 SC_SlotWrite := SlotWrite;
161
162 SC_OUT_Pusher_1 := OUT_Pusher_1;
163
164 SC_OUT_Clasificador_FnF_Plus := OUT_Pop_Up_Wheel_Sorter_4_Plus;
165 SC_OUT_Clasificador_FnF_Right := OUT_Pop_Up_Wheel_Sorter_4_Right;
166
167 SC_OUT_Clasificador_NF_Plus := OUT_Pop_Up_Wheel_Sorter_1_Plus;
168 SC_OUT_Clasificador_NF_Right := OUT_Pop_Up_Wheel_Sorter_1_Right;
169
170 SC_OUT_Clasificador_F_Plus := OUT_Pop_Up_Wheel_Sorter_3_Plus;
171 SC_OUT_Clasificador_F_Left := OUT_Pop_Up_Wheel_Sorter_3_Left;
172
173 SC_IN_Medicion_Tamano_F := IN_Light_Array_Emitter_1_Value;
174 SC_IN_Medicion_Tamano_NF := IN_Light_Array_Emitter_2_Value;
175
176 SC_IN_Sensor_Tamano_F := IN_Diffuse_Sensor_8;
177 SC_IN_Sensor_Tamano_NF := IN_Diffuse_Sensor_3;
178
179 SC_IN_Sensor_Tamano :=
180
181     SC_IN_Sensor_Tamano_NF OR
182     SC_IN_Sensor_Tamano_F;
183
184
185 (* ===== *)
186 (* PICK & PLACE *)
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Visual

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1|      10|      20|      30|      40|      50|      60|      70|      80|      90|     100|     110|
187  (* ===== *)
188
189  SC_IN_Pieza_Azul_PP := IN_Diffuse_Sensor_15;
190  SC_IN_Pieza_Verde_PP := IN_Diffuse_Sensor_12;
191
192  SC_IN_Base_Azul := IN_Right_Positioner_4_Clamped;
193  SC_IN_Base_Verde := IN_Right_Positioner_2_Clamped;
194
195  SC_IN_PP_Azul_Detected := IN_TwoAxis_PickPlace_2_Detected;
196  SC_IN_PP_Verde_Detected := IN_TwoAxis_PickPlace_1_Detected;
197
198  SC_IN_PP_Azul_Moving_X := IN_TwoAxis_PickPlace_2_Moving_X;
199  SC_IN_PP_Azul_Moving_Z := IN_TwoAxis_PickPlace_2_Moving_Z;
200  SC_IN_PP_Verde_Moving_X := IN_TwoAxis_PickPlace_1_Moving_X;
201  SC_IN_PP_Verde_Moving_Z := IN_TwoAxis_PickPlace_1_Moving_Z;
202
203  SC_OUT_Right_Positioner_2_Clamp := OUT_Right_Positioner_2_Clamp;
204  SC_OUT_Right_Positioner_2_Raise := OUT_Right_Positioner_2_Raise;
205  SC_OUT_Right_Positioner_4_Clamp := OUT_Right_Positioner_4_Clamp;
206  SC_OUT_Right_Positioner_4_Raise := OUT_Right_Positioner_4_Raise;
207
208  SC_OUT_BeltConveyor_4_2 := OUT_BeltConveyor_4_2;
209  SC_OUT_BeltConveyor_4_7 := OUT_BeltConveyor_4_7;
210
211
212  (* ===== *)
213  (* ALMACENAMIENTO / RACKS *)
214  (* ===== *)
215
216  SC_Position_Azul := Position;
217  SC_Position_Verde := Position_V;
218  SC_Position_F := Position_F;
219  SC_Position_NF := Position_NF;
220
221  SC_Rack_Azul_Max := 54;
222  SC_Rack_Verde_Max := 54;
223  SC_Rack_F_Max := 54;
224  SC_Rack_NF_Max := 54;
225
226  SC_Paso_Azul := Paso;
227  SC_Paso_Verde := Paso_V;
228  SC_Paso_F := Paso_F;
229  SC_Paso_NF := Paso_NF;
230
231  SC_Rack_Azul_Lleno := Position >= 54;
232  SC_Rack_Verde_Lleno := Position_V >= 54;
233  SC_Rack_F_Lleno := Position_F >= 54;
234  SC_Rack_NF_Lleno := Position_NF >= 54;
235
236  SC_IN_Entrada_Azul_Roller := IN_Diffuse_Sensor_24;
237  SC_IN_Entrada_Verde_Roller := IN_Diffuse_Sensor_4;
238  SC_IN_Entrada_F_Roller := IN_Diffuse_Sensor_23;
239  SC_IN_Entrada_NF_Roller := IN_Diffuse_Sensor_21;
240
241  SC_IN_Entrada_Azul_Stacker := IN_Diffuse_Sensor_25;
242  SC_IN_Entrada_Verde_Stacker := IN_Diffuse_Sensor_26;
243  SC_IN_Entrada_F_Stacker := IN_Diffuse_Sensor_22;
244  SC_IN_Entrada_NF_Stacker := IN_Diffuse_Sensor_20;
245
246  SC_OUT_StCr_1_TargetPosition := OUT_StCr_1_TargetPosition;
247  SC_OUT_StCr_2_TargetPosition := OUT_StCr_2_TargetPosition;
248  SC_OUT_StCr_3_TargetPosition := OUT_StCr_3_TargetPosition;
249  SC_OUT_StCr_4_TargetPosition := OUT_StCr_4_TargetPosition;
250
251
252  (* ===== *)
253  (* STACKER CRANES *)
254  (* ===== *)
255
256  SC_OUT_Stacker_Crane_1_Lift := OUT_Stacker_Crane_1_Lift;
257  SC_OUT_Stacker_Crane_1_Left := OUT_Stacker_Crane_1_Left;
258  SC_OUT_Stacker_Crane_1_Right := OUT_Stacker_Crane_1_Right;
259
260  SC_OUT_Stacker_Crane_2_Lift := OUT_Stacker_Crane_2_Lift;
261  SC_OUT_Stacker_Crane_2_Left := OUT_Stacker_Crane_2_Left;
262  SC_OUT_Stacker_Crane_2_Right := OUT_Stacker_Crane_2_Right;
263
264  SC_OUT_Stacker_Crane_3_Lift := OUT_Stacker_Crane_3_Lift;
265  SC_OUT_Stacker_Crane_3_Left := OUT_Stacker_Crane_3_Left;
266  SC_OUT_Stacker_Crane_3_Right := OUT_Stacker_Crane_3_Right;
267
268  SC_OUT_Stacker_Crane_4_Lift := OUT_Stacker_Crane_4_Lift;
269  SC_OUT_Stacker_Crane_4_Left := OUT_Stacker_Crane_4_Left;
270  SC_OUT_Stacker_Crane_4_Right := OUT_Stacker_Crane_4_Right;
271
272  SC_Stacker_Azul_Activo :=
273      OUT_Stacker_Crane_1_Lift OR
274      OUT_Stacker_Crane_1_Left OR
275      OUT_Stacker_Crane_1_Right;
276
277  SC_Stacker_NF_Activo :=
278      OUT_Stacker_Crane_2_Lift OR
279      OUT_Stacker_Crane_2_Left OR
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Visual

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1|      10|      20|      30|      40|      50|      60|      70|      80|      90|      100|      110|

280          OUT_Stacker_Crane_2_Right;
281
282 SC_Stacker_F_Activo :=
283     OUT_Stacker_Crane_3_Lift OR
284     OUT_Stacker_Crane_3_Left OR
285     OUT_Stacker_Crane_3_Right;
286
287 SC_Stacker_Verde_Activo :=
288     OUT_Stacker_Crane_4_Lift OR
289     OUT_Stacker_Crane_4_Left OR
290     OUT_Stacker_Crane_4_Right;
291
292
293 (* ===== *)
294 (* CONTADORES INTERNOS DE PALLETIZADO *)
295 (* ===== *)
296
297 SC_Contador_verdes := Contador_verdes;
298 SC_Contador_azules := Contador_azules;
299 SC_Contador_gris := Contador_gris;
300 SC_Contador_PF := Contador_PF;
301 SC_Contador_P_nF := Contador_P_nF;
302 SC_Contador_NF_G := Contador_NF_G;
303 SC_Contador_F_G := Contador_F_G;
```